Marketing Campaign Budget Allocation Using Mixed Integer Linear Programming

1. Problem Description

Marketing managers often need to allocate a limited budget across multiple advertising campaigns operating in different channels such as social media, search, email, and television. Because budgets are constrained, decision makers must determine which campaigns to select in order to allocate resources efficiently. Selecting too many campaigns may exceed the available budget, while selecting too few can reduce reach and overall return on investment.

This project formulates the problem as a mathematical optimization model that uses campaign level data such as cost, ROI, reach, and marketing channel to identify the set of campaigns that maximizes expected return while satisfying operational constraints. The resulting solution represents a campaign portfolio that balances return, cost, and reach.

2. Model Classification

The campaign allocation problem considered in this project is formulated as a Mixed Integer Linear Programming (MILP) model. In this framework, decision variables are binary and represent whether a specific marketing campaign is selected. For each campaign i, the variable takes the value 1 if the campaign is chosen and 0 otherwise. Because these variables are restricted to integer values, the problem belongs to the class of mixed integer optimization models rather than purely continuous linear programs.

The model uses a linear objective function that maximizes the total expected return of selected campaigns while optionally penalizing higher risk campaigns, subject to linear constraints reflecting practical marketing limitations such as budget capacity, minimum audience reach, and limits on the number of campaigns per marketing channel.

Structurally, this problem is closely related to the classical 0–1 knapsack problem, where items with different values and costs must be selected under a capacity constraint. However, additional constraints such as channel limits and reach requirements extend the model beyond the basic knapsack formulation.

3. Sets and Indices

To formally represent the marketing campaign allocation problem, several index sets are defined. These sets describe the elements involved in the optimization model.

Let:

$$i \in I$$

be the set of all available marketing campaigns in the dataset.

Each element i represents a single campaign that can potentially be selected as part of the marketing plan.

Additionally, marketing campaigns belong to different advertising channels such as social media, search advertising, email marketing, or television.

Let:

$$k \in K$$

denote the set of marketing channels.

Each campaign $i$belongs to exactly one channel k. The set of campaigns that belong to channel $k$is denoted as:

$$C_k \subseteq I$$

These sets allow the model to impose restrictions on how many campaigns can be selected from each channel, ensuring a balanced marketing strategy.

4. Parameters

The optimization model uses several parameters derived from the marketing campaign dataset. These parameters represent fixed input values that characterize each campaign.

For each campaign $i \in I$, the following parameters are defined:

$$Cost_i$$

The acquisition cost of campaign i. This parameter represents the financial investment required to run the campaign.

$$ROI_i$$

The expected return on investment generated by campaign i. This value is obtained directly from the dataset and represents the predicted marketing return associated with the campaign.

$$Reach_i$$

The expected audience reach of campaign i. This value represents the potential number of users exposed to the campaign. In the dataset, reach can be approximated using variables such as impressions or engagement indicators.

$$Risk_i$$

The risk level associated with campaign i. This parameter reflects uncertainty in the expected performance of the campaign. A proxy measure may be derived from campaign performance variability or conversion related metrics.

In addition to campaign level parameters, the model includes several global parameters that describe operational constraints.

$$B$$

Total available marketing budget. This parameter limits the maximum total cost of the selected campaigns.

$$R_{min}$$

Minimum required audience reach. The selected campaigns must collectively achieve at least this level of exposure.

$$L_k$$

Maximum number of campaigns that can be selected from marketing channel k. This constraint ensures that the marketing plan maintains diversity across advertising channels.

$$\lambda$$

Risk penalty parameter that determines how strongly high risk campaigns are penalized in the objective function.

5. Decision Variables

The optimization model determines whether each campaign should be included in the final marketing portfolio.

For each campaign $i \in I$, the following binary decision variable is defined:

$$x_i = \begin{cases} 1 & \text{if campaign } i \text{ is selected} \\ 0 & \text{otherwise} \end{cases}$$

The value of $x_i$is determined by the optimization model. If $x_i = 1$, the campaign is included in the marketing plan and contributes to the total cost, reach, and return. If $x_i = 0$, the campaign is excluded and has no effect on the solution.

Because these decision variables are binary, the model belongs to the class of mixed integer linear programming (MILP) problems.

The set of decision variables collectively determines the portfolio of marketing campaigns that will be implemented within the available budget and operational constraints.

6. Objective Function

The primary objective of the optimization model is to select the combination of marketing campaigns that maximizes the total expected return while considering potential campaign risks.

Each campaign generates an expected return on investment (ROI), but campaigns with higher uncertainty or performance variability may also introduce additional risk into the marketing strategy.

To balance these two factors, the objective function includes both a return component and a risk penalty component.

The objective function of the model is defined as:

$$\max \sum_{i \in I} ROI_i x_i - \lambda \sum_{i \in I} Risk_i x_i$$

Total expected return generated by the

Subtracts a penalty proportional to campaign risk

where:

- ✓ ROI_irepresents the expected return generated by campaign i
- ✓ $Risk_i$represents the risk level associated with campaign i
- ✓ x_i is the binary decision variable indicating whether campaign $i$is selected
- ✓ λ is a non negative parameter that controls the importance of the risk penalty

By adjusting the parameter λ, decision makers can control how strongly risk influences the campaign selection process.

If $\lambda = 0$, the model focuses entirely on maximizing return. As λ increases, the model becomes more sensitive to the risk and tends to select campaigns with lower risk levels.

7. Model Constraints

7.1 Budget Constraint

The total cost of the selected campaigns cannot exceed the available marketing budget.

$$\sum_{i \in I} Cost_i x_i \leq B$$

where:

- ✓ $Cost_i$is the acquisition cost of campaign i
- ✓ Brepresents the total available marketing budget

This constraint ensures that the selected campaign portfolio remains financially feasible.

7.2 Minimum Reach Constraint

Marketing strategies typically aim to achieve a minimum level of audience exposure. Therefore, the selected campaigns must collectively generate at least a predefined level of audience reach.

$$\sum_{i \in I} Reac\, h_i x_i \geq R_{min}$$

where:

- ✓ $Reach_i$represents the audience reach of campaign i
- ✓ $R_{min}$is the minimum required total reach

This constraint guarantees that the selected campaigns provide sufficient visibility in the market.

7.3 Channel Capacity Constraint

To maintain a balanced marketing strategy, organizations may limit the number of campaigns selected from each marketing channel.

For each channel $k \in K$, the following constraint is imposed:

$$\sum_{i \in C_k} x_i \leq L_k$$

where:

- ✓ $C_k$is the set of campaigns belonging to channel k
- ✓ $L_k$represents the maximum number of campaigns allowed for that channel

This constraint prevents the model from concentrating all selected campaigns within a single marketing channel and encourages diversification across different advertising platforms.

7.4 Binary Decision Constraint

The decision variables are binary, meaning that each campaign can either be selected or not selected.

$$x_i \in \{0,1\} \forall i \in I$$

This constraint ensures that campaigns are either fully included in the marketing plan or completely excluded.

8. Model Assumptions

To formulate the marketing campaign allocation problem as a tractable optimization model, several simplifying assumptions are made.

First, campaign returns are assumed to be independent and additive, meaning the total return equals the sum of returns from the selected campaigns.

Second, campaign costs are treated as fixed and known, even though they may vary in real markets.

Third, the expected reach of each campaign is assumed to remain constant during the planning period and is estimated using engagement related dataset metrics.

The model also considers a single period planning horizon, ignoring long term marketing effects. Finally, campaign risk is approximated using available performance indicators, since a direct risk metric is not provided in the dataset.

9. Model Limitations

Although the proposed optimization model offers a structured framework for campaign selection, it has several limitations. The model does not account for interactions between marketing channels, even though campaigns in different channels may reinforce each other in practice.

Another limitation is that the model focuses mainly on short term performance indicators such as return on investment and campaign reach, while long term outcomes like brand awareness and customer loyalty are not considered.

In addition, campaign performance is estimated using historical data, which may not fully reflect future market conditions. The model also treats campaigns as binary decisions, meaning they are either selected or not, while in reality companies may allocate partial budgets or adjust campaign intensity levels.

Despite these limitations, the model still provides a useful decision support tool for evaluating campaign portfolios under budget constraints.

10. Expected Computational Complexity

The marketing campaign allocation problem formulated in this project belongs to the class of Mixed Integer Linear Programming (MILP) problems. Optimization problems with binary decision variables are generally computationally challenging, since the number of possible solutions grows exponentially with the number of campaigns.

More specifically, the structure of the model is closely related to the classical 0–1 knapsack problem, which is known to be NP hard. When additional constraints such as reach requirements and channel limits are included, the resulting problem becomes an extended combinatorial optimization problem.

The computational complexity of the model primarily depends on several factors:

- ✓ the number of available campaigns in the dataset
- ✓ the number of constraints included in the formulation
- ✓ the solver used to compute the optimal solution

If the number of campaigns becomes very large, solving the optimization model may require significant computational effort. However, modern optimization solvers such as Gurobi or other MILP solvers are capable of solving medium scale problems efficiently using advanced branch and bound and cutting plane techniques.

In this project, computational experiments will be conducted using subsets of the dataset to ensure that the model can be solved within a reasonable runtime while still providing meaningful insights into marketing campaign selection strategies.